

Automation Exercise — QA Testing Project

A comprehensive Quality Assurance project covering Manual Testing, API Testing, Test Automation, and Database Testing for the [Automation Exercise](#) web application, with project management tracked via Jira.

Project Overview

This project was developed as a full-cycle QA exercise, simulating a real-world testing workflow. It includes:

- Manual UI test cases covering core user flows
 - API testing for key endpoints
 - Automated test scripts using Selenium + Java + TestNG
 - Database validation queries using the Sakila sample database
 - Test planning and defect tracking via Jira
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Project Structure

— Manual/	# Manual test cases and test scenarios
— APIs/	# API test cases and collections
— Database/	# SQL queries for Sakila DB validation
— Our test Plan - Group1.docx	# Official test plan document

Tech Stack & Tools

Area	Tools / Technologies
Automation	Selenium WebDriver, Java, TestNG
Manual Testing	Excel / Google Sheets, Test Case Design
API Testing	Postman / REST API validation
Database Testing	MySQL, Sakila Database
Test Management	Jira (Scrum Board)
Version Control	Git & GitHub

Project Resources

Resource	Link
Application Under Test	automationexercise.com
Sakila Database	Download from MySQL
Jira Board	View Backlog (<i>requires Jira login — contact the team for access</i>)
Project Artifacts (Drive)	View on Google Drive (<i>publicly accessible — no login required</i>)

Testing Scope

Manual Testing

- Test cases for UI flows: Registration, Login, Product Search, Cart, Checkout, Contact Form
- Covered positive, negative, and edge cases
- Linked to Jira user stories and tracked for pass/fail status

API Testing

- Validated REST API endpoints provided by the site
- Verified status codes, response bodies, and data integrity

Automated Testing (Selenium + Java + TestNG)

- Page Object Model (POM) design pattern
- Data-driven test execution
- Assertions and reporting via TestNG

Database Testing (Sakila DB)

- SQL queries to validate data at the database layer
 - Covered SELECT, JOIN, aggregation, and filter scenarios
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Test Plan

The full test plan is available in the repository:

[Our test Plan - Group1.docx](#)

Team

This project was collaboratively developed by a QA team as part of a structured testing exercise.

Contact

For access to Jira or any restricted resources, please reach out to a project contributor via GitHub.